23CCE385 – OPEN LAB 2
DRIVE BY WIRELESS SYSTEM

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Board Used

For Both Driver-side and Vehicle Side

ESP32-WROOM-32 Development Board

- Dual-core 32-bit MCU with integrated Wi-Fi & Bluetooth
- Ultra-low power consumption
- Multiple communication interfaces
- Suitable for real-time sensor data collection & wireless communication
- Supports on-board battery-powered operation

1. **Driver-Side ESP32**

   Purpose: read human input

2. **Vehicle-Side ESP32**

   Purpose: Actuate wheels

Sensors Used

Steering Input

Rotary encoder connected to steering wheel shaft.
Encoder output: **A, B channel pulses**
ESP32 converts to **steering angle**.

Pedal Inputs

Two potentiometers

- Pot 1 → accelerator
- Pot 2 → brake

ESP32 reads **analog values (0–4095)** using ADC.

Actuators Used

Front Wheel Steering
Servo motor

- fixed to steering mechanism.
- Receives **angle commands** from Driver-Side via wireless.
- Maps steering angle → **Servo PWM**

Rear Wheel Drive

DC motor with motor driver (L298N).
Accelerator value → **PWM duty cycle**
Brake:
    Active braking → reverse PWM or simply stop motor.

Interfacing of the Sensors:

GPIO (Digital Inputs)

- Used for **rotary encoder**
- Reads **quadrature pulses** from channels A and B
- Connected to **interrupt-capable pins** on ESP32

ADC (Analog Inputs)

- Used for **accelerator and brake potentiometers**
- ESP32 ADC converts **0–3.3V → 12-bit digital value (0–4095)**
- Continuous sampling at **50–100 Hz**

VEHICLE SIDE:

Module	Pin / Signal	ESP32 GPIO / Supply
SG90 Servo	PWM (Signal)	GPIO 33
	VCC	5 V
	GND	GND

Module	Pin / Signal	ESP32 GPIO / Supply
L298 Motor Driver	ENA	GPIO 25
	IN1	GPIO 26
	IN2	GPIO 27
	ENB	GPIO 14
	IN3	GPIO 12
	IN4	GPIO 13
	GND	GND
	VS (Motor Supply)	7.4 V Battery
	VSS (Logic Supply)	5 V

THANK YOU !!